

Machine Problem 01: Stippled Star Frame (GL_LINE_LOOP)

- **Objective:** Practice combining GL_LINE_LOOP, custom line thickness (glLineWidth), and stipple pattern masks (glLineStipple) to draw a geometric star frame.
- **Task:** Write an OpenGL program that calculates or defines the vertices of a 5pointed star. Render the outer boundary using GL_LINE_LOOP with a line width of 3.0f and a custom stipple pattern (0x0F0F) scaled by a repeat factor of 2.



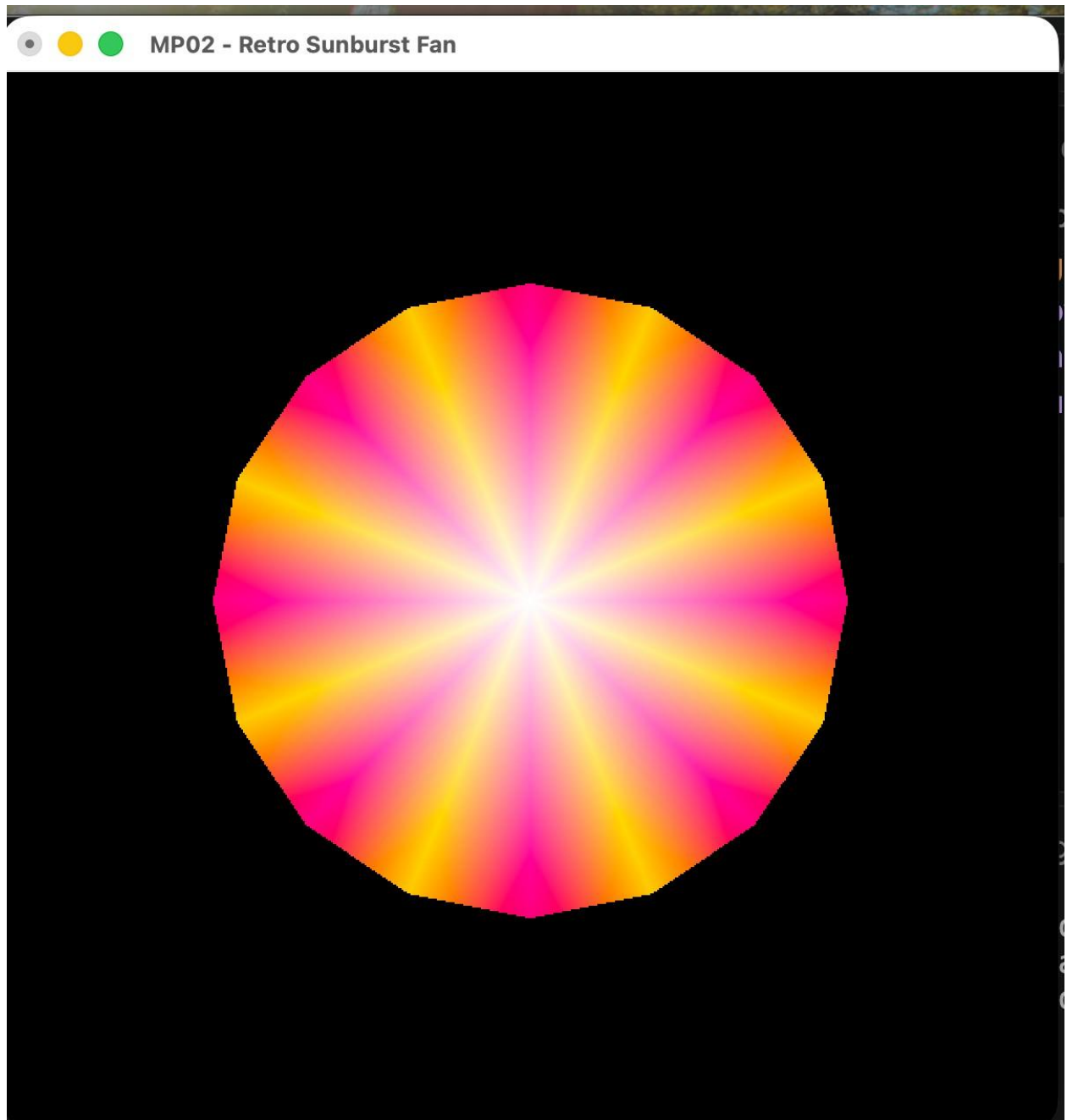
Output:



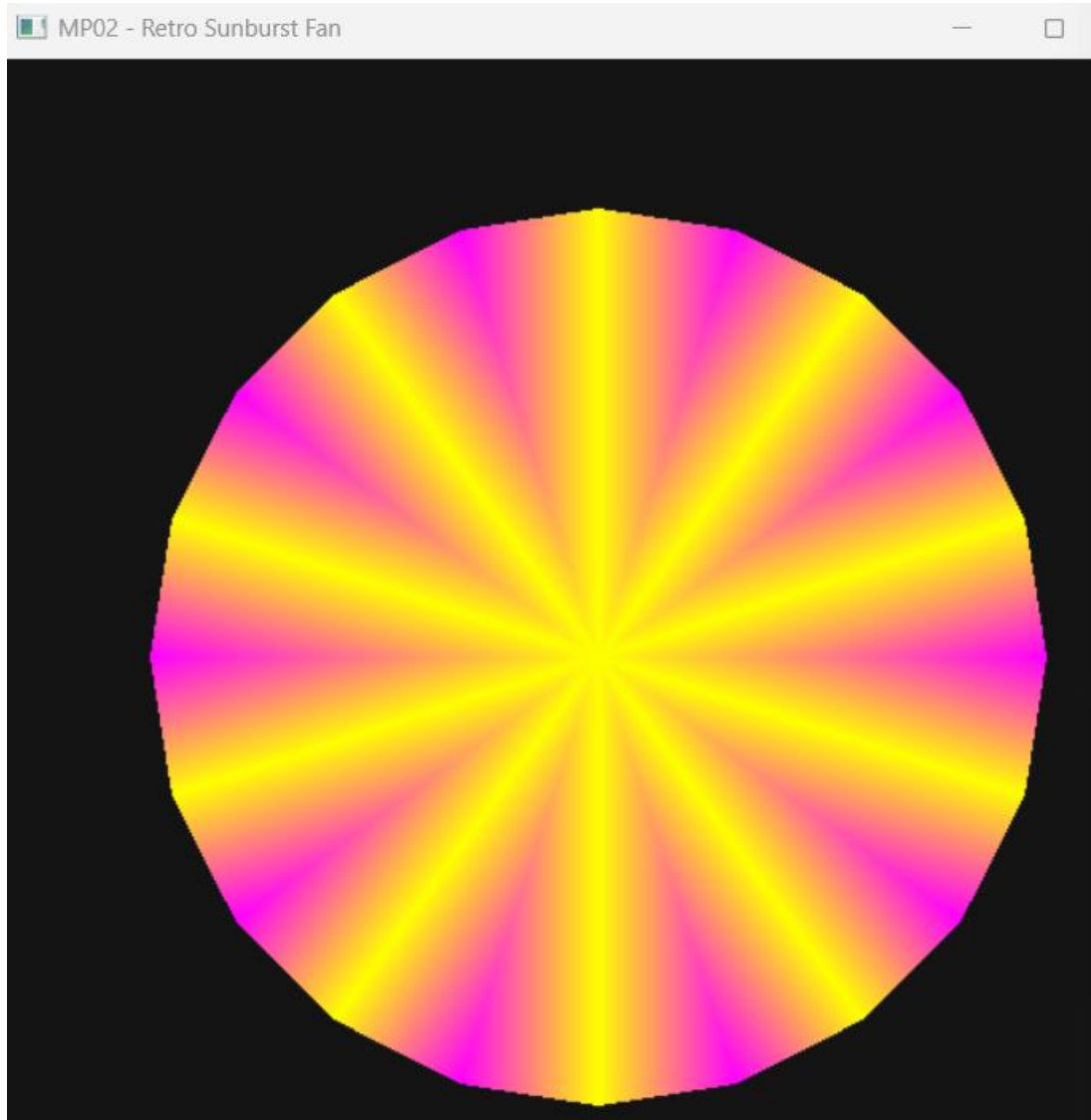
Machine Problem 02: Retro Sunburst Fan (GL_TRIANGLE_FAN)

- **Objective:** Master GL_TRIANGLE_FAN by creating a multi-segment radial burst with alternating vertex colors.
- **Task:** Create a full 360-degree radial sunburst from a central hub vertex using GL_TRIANGLE_FAN. Generate at least 16 outer segments procedurally via a loop, alternating the slice colors between bright magenta and yellow.

Module 2 – Machine Problem



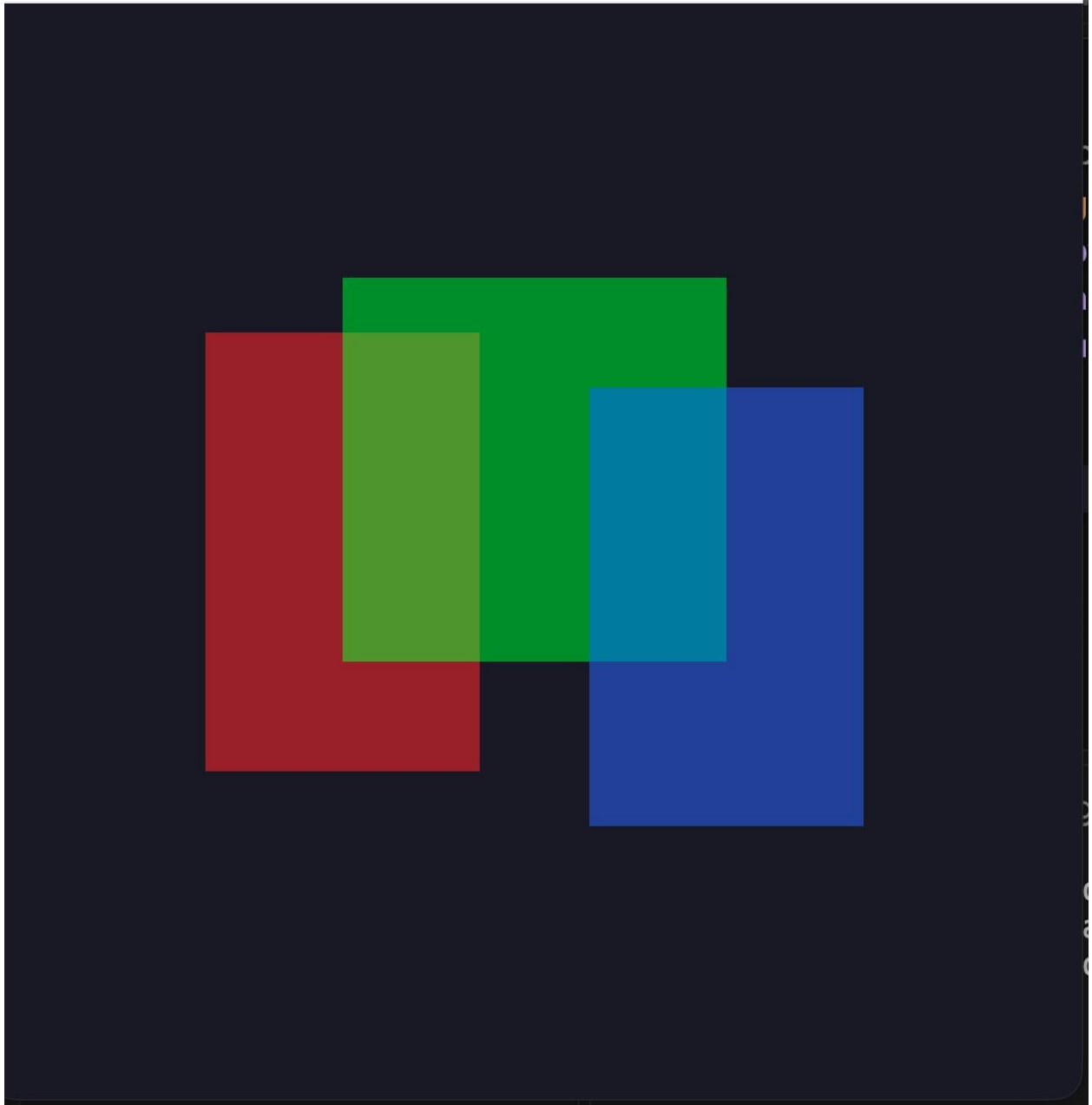
Output:



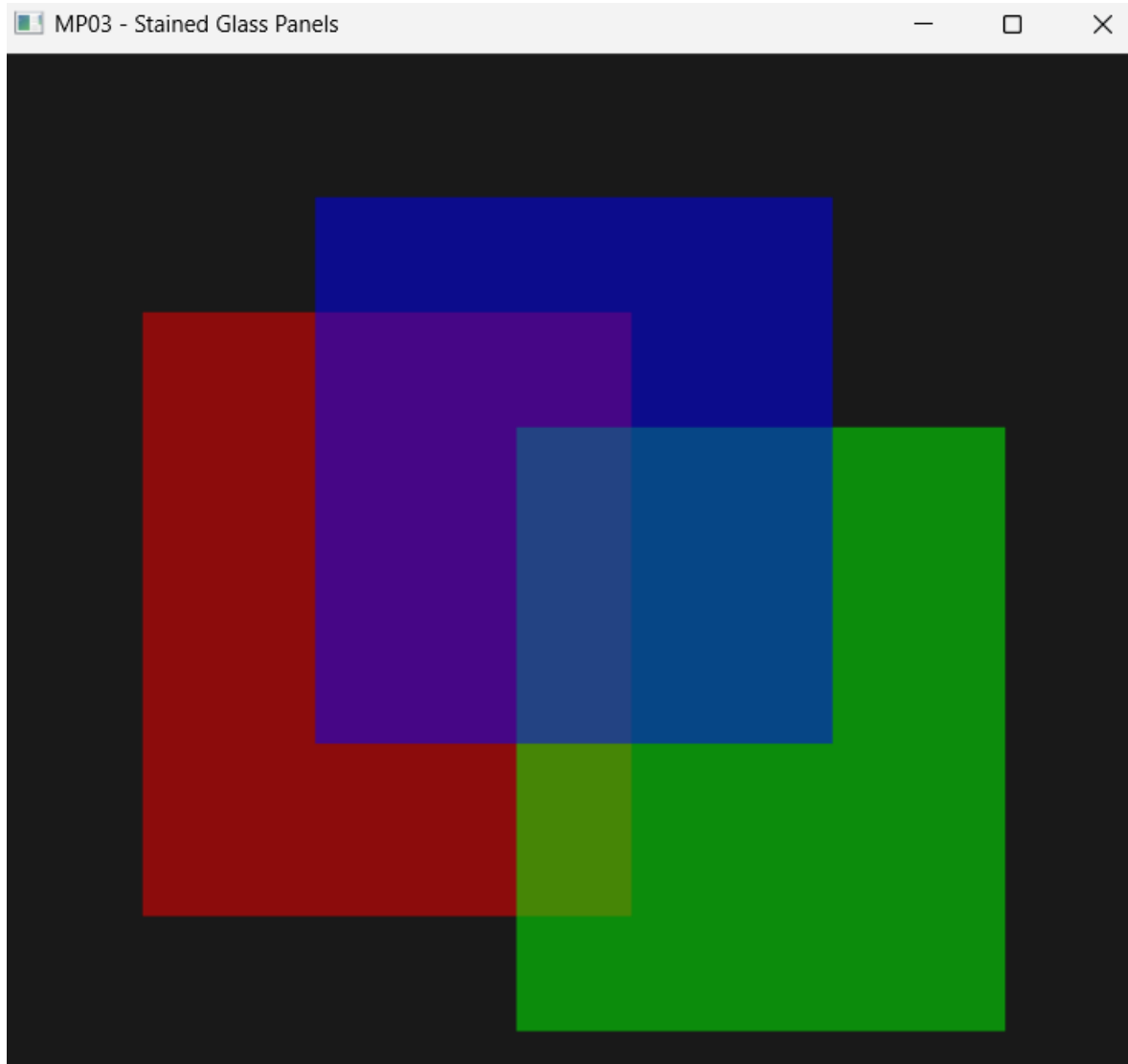
Machine Problem 03: Stained Glass Panels (GL_QUADS & Alpha Blending)

- **Objective:** Practice `glColor4f` combined with `glEnable(GL_BLEND)` to render overlapping translucent rectangular panels.
- **Task:** Write a program that draws three intersecting quadrilaterals (`GL_QUADS`) at different positions. Assign each quad a distinct vibrant color with an alpha value of 0.5f so that their overlapping intersections create mixed secondary hues.

MP03 - Stained Glass Panels



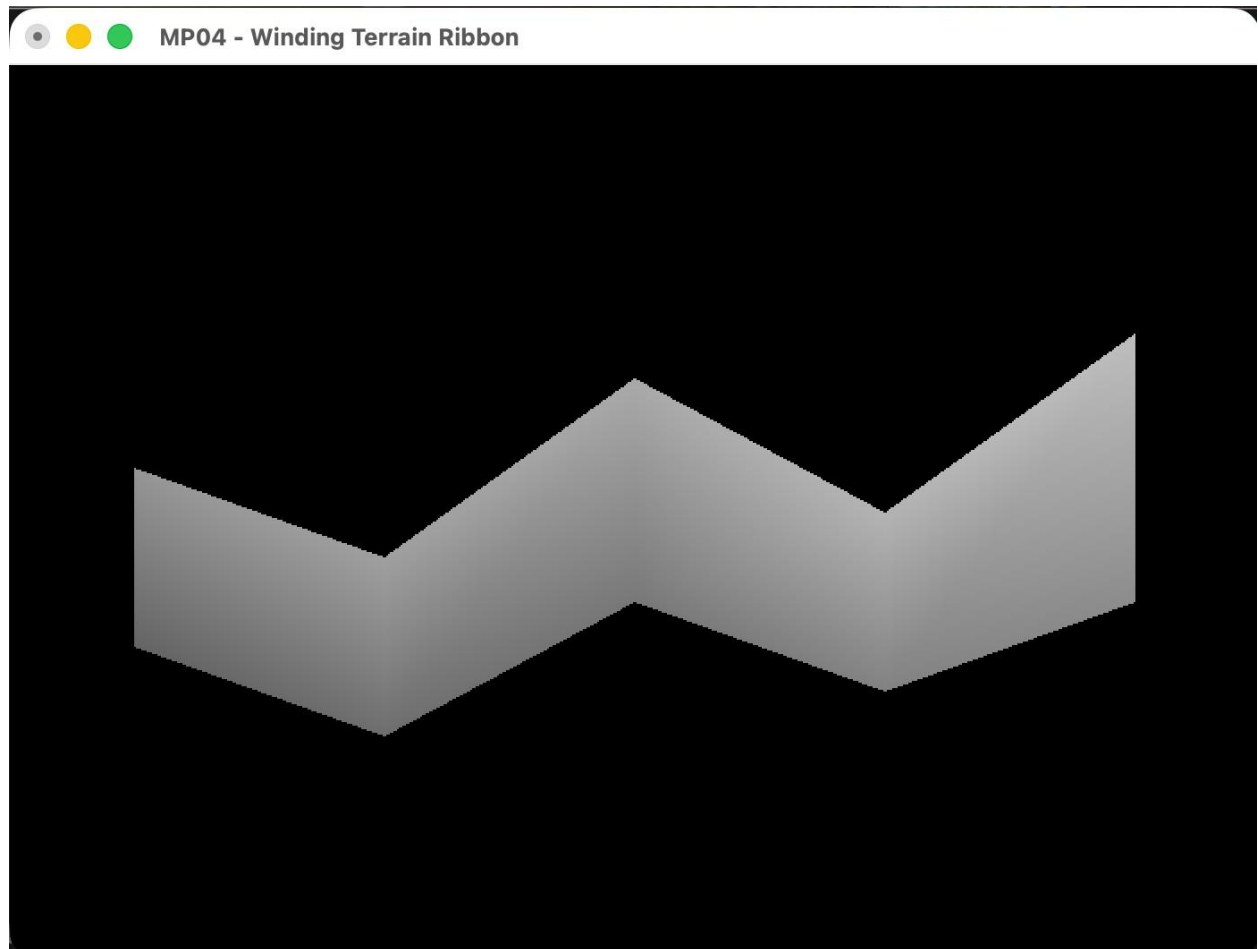
Output:



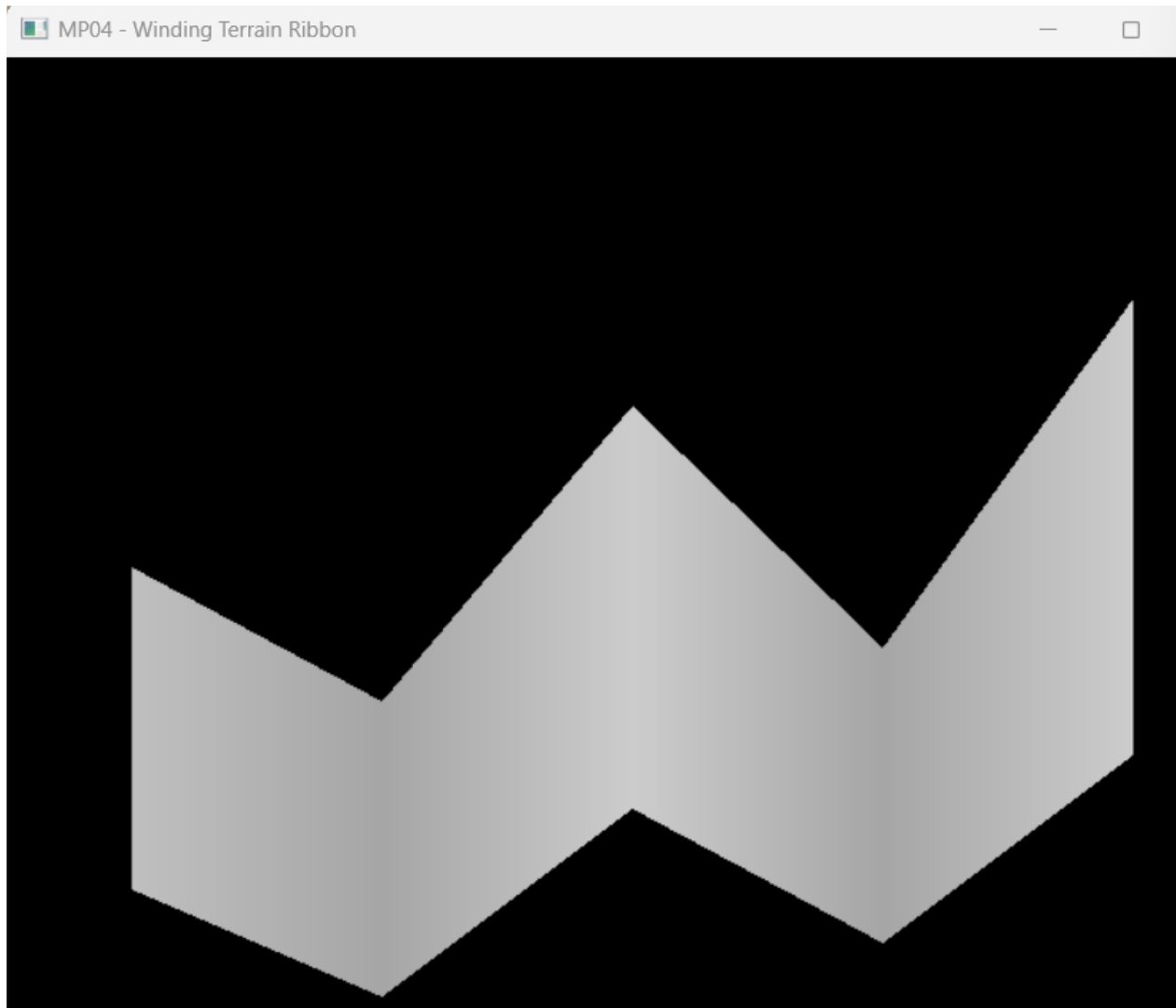
Machine Problem 04: Winding Terrain Ribbon (GL_QUAD_STRIP)

- **Objective:** Practice the alternating bottom/top vertex pairing rules of GL_QUAD_STRIP to create a continuous ribbon.
- **Task:** Use GL_QUAD_STRIP with at least 5 pairs of vertices to construct an undulating ribbon across the screen. Apply per-pair colors so each segment shifts in tone, simulating a curving pathway or ribbon.

Module 2 – Machine Problem



Output:



Machine Problem 05: Architectural House Scene (Capstone)

- **Objective:** Combine multiple primitive types (GL_QUADS, GL_TRIANGLES, GL_POLYGON) into a cohesive composite scene.
- **Task:** Create a scene depicting a house (walls made of GL_QUADS, a roof made of GL_TRIANGLES, a door using quads, and a lawn/ground layer using GL_POLYGON).

Module 2 – Machine Problem



Output:

Module 2 – Machine Problem

